

SYSTEM IN IOT SYSTEMS

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Introduction

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Project Overview

- ✂ An Intrusion Detection System (IDS) monitors IoT networks for malicious activity
- ✂ Hardware platform: RFID reader (UID source), ESP32 microcontroller, Raspberry Pi
- ✂ ML model trained on NSL-KDD dataset using Random Forest classifier
- ✂ IDS deployed on Raspberry Pi for real-time intrusion detection

Objectives

- ✂ Detect and classify network intrusions in IoT environments
- ✂ Deploy a lightweight ML model on resource-constrained hardware

Problem Statement

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The Problem

- ✂ IoT devices are increasingly targeted by cyber attacks due to limited built-in security
- ✂ DoS attacks overwhelm IoT nodes, disrupting communication and availability
- ✂ Replay attacks retransmit captured packets to gain unauthorized access
- ✂ RFID-based systems are vulnerable to UID spoofing and unauthorized scanning

Why This Problem Matters

- ✂ Billions of IoT devices deployed globally — security breaches have critical consequences
- ✂ Traditional intrusion detection systems are too heavy for IoT hardware
- ✂ Real-time detection is essential to minimize damage from ongoing attacks
- ✂ A lightweight, ML-driven IDS is needed that can run on edge devices like Raspberry Pi

Literature Review

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Summary of Existing Work

- ✂ Intrusion detection in IoT has been widely studied using signature-based and anomaly-based approaches
- ✂ Signature-based methods (e.g., Snort) are fast but fail to detect novel attacks
- ✂ ML-based IDS using Random Forest, SVM, and Decision Trees show high accuracy on benchmark datasets

Limitations & Gaps in Existing Solutions

- ✂ Most existing IDS solutions are designed for traditional networks, not IoT-specific architectures
- ✂ High computational overhead makes deep learning models impractical on edge devices
- ✂ Lack of real-time web interface for monitoring in embedded IDS deployments

Work Plan

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Week	Work Done	Status
Week 1	Studied IDS concepts and implementation on Raspberry Pi.	Completed
Week 2	Researched ML-driven IDS approaches; analysed case studies and benchmark datasets	Completed
Week 3	Understood IoT system architecture, workflow, and vulnerabilities in IoT layers	Completed
Week 4	Research paper analysis on IDS in IoT	Completed
Week 5	Thorough study of selected research papers	Completed
Week 6		
Week 7		
Week 8		
Week 9	Error fixation of code while building firmware on ESP32	Completed
Week 10	Enabled WiFi mode on ESP32; completed GPIO pin communication with ESP32	Completed

Work Plan

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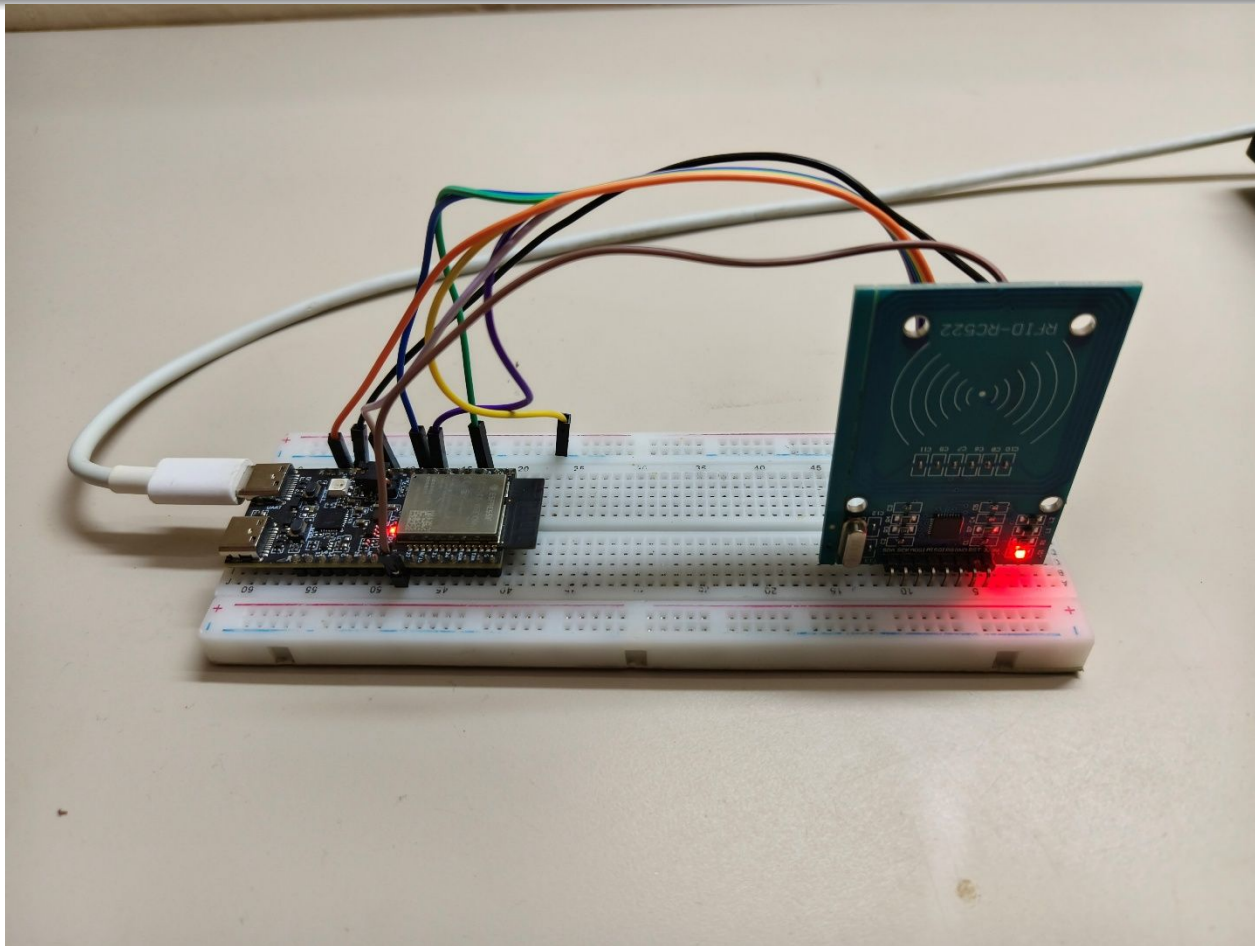
Week	Work Done	Status
Week 11	Establishing connection between ESP32 and Raspberry pi.	Completed
Week 12	Resolving errors and final testing of project.	Completed
Week 13	PPT , Report	Completed

Individual Contribution

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Member Name	Enrollment No.	Contribution
Ranjan Mishra	231B258	Model Development, and Research in light weight models in RPI for IDS.
Rashmi Dubey	231b260	IoT architecture study, Possible vulnerabilities at different layers of IOT Architecture,
Rishabh Jain	231B264	Hardware integration of rfid, raspberry pi, and esp32c5

Work Done So Far



Work Done So Far — System Architecture

IoT System Architecture

- ✂ Perception Layer: RFID reader captures UID of IoT tags
- ✂ Network Layer: ESP32-C5 reads RFID data and transmits it via WiFi using MQTT protocol to Raspberry Pi
- ✂ Application Layer: Raspberry Pi runs MQTT broker and IDS logic to analyze incoming RFID data and detect attacks

Attack Flow (Behavioral Diagram)

- ✂ Unauthorized Access: If the scanned UID does not match any entry in the authorized users list, the status is marked as "ALERT"
- ✂ Replay Attack: Backend maintains a frequency of scans for every uid, and if a card is scanned more than 10 times in rapid succession, alert is sent.

Data Flow

- ✂ RFID Reader → ESP32-C5 (WiFi) → Raspberry Pi (IDS + ML Model) → Alert/Log

Work Done So Far — Hardware Setup

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Hardware Components

- ✂ RFID Module: Reads tag UIDs; acts as data source for the IoT node
- ✂ ESP32-C5 Microcontroller: Handles WiFi connectivity and GPIO communication
 - 📁 WiFi mode successfully configured on ESP32-C5
 - 📁 GPIO pin communication established
- ✂ Raspberry Pi: Central IDS node running ML model for real-time detection

Implementation Progress

- ✂ ESP32-C5 firmware coded and debugged — errors resolved
- ✂ WiFi mode on ESP32-C5 operational
- ✂ GPIO communication between RFID and ESP32-C5 verified
- ✂ System is correctly identifying the possible attacks and threats.

Work Done So Far — ML Model

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- **Model Training Details**
- **Random Forest Model (NSL-KDD)**
 - Trained using ~100 decision trees
 - Applied **SMOTE** to handle class imbalance
 - Achieved **~99% accuracy**
 - Strong performance in detecting known attack patterns
- **XGBoost Model (CICIoT2023)**
 - Trained on IoT-specific network traffic
 - Captures complex patterns in modern cyber-attacks
 - Optimized for high performance and scalability
- **Training Strategy**
 - Models trained **independently on individual datasets**
 - Dataset-specific learning improves specialization

Work Done So Far — Dataset Details

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- **Datasets Used for Model Training & Evaluation**
- **NSL-KDD Dataset**
 - Improved version of KDD'99 (removes redundancy)
 - Contains labeled network traffic (normal + attack types)
 - Useful for benchmarking intrusion detection systems
- **CICIoT2023 Dataset**
 - Modern IoT-focused dataset with realistic traffic scenarios
 - Includes diverse attack categories (DDoS, DoS, spoofing, etc.)
 - Captures real-world IoT communication behavior
- **MQTT Dataset**
 - Focused on MQTT protocol used in IoT systems
 - Includes attacks like flooding, malformed packets, and unauthorized access
 - Helps model protocol-specific intrusion patterns
- **Why Multiple Datasets?**
- Ensures **diversity of attack patterns**
- Improves **model generalization**
- Covers both **traditional networks + IoT environments**

Work To Be Done

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Develop Ensemble Models

- ▣ Combine Random Forest + XGBoost predictions
 - ▣ Improve robustness and overall accuracy
- Apply **GAN (Generative Adversarial Networks)**
 - ▣ Generate synthetic attack data
 - ▣ Enhance dataset diversity and handle imbalance
- Move towards **Unified IDS for IoT + Traditional Networks**
- ✂ Deploy trained Random Forest ML model on Raspberry Pi
 - 📁 Validate real-time inference on live RFID/ESP32-C5 traffic
- ✂ End-to-end integration testing of the full system
 - 📁 RFID → ESP32-C5 → Raspberry Pi pipeline with live IDS response

Conclusion

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Summary of Progress

- ✂ Completed in-depth study of IDS concepts, IoT architecture, and ML-based detection approaches
- ✂ Analysed relevant research papers and identified gaps addressed by this project
- ✂ Hardware setup complete: ESP32-C5 WiFi mode and GPIO communication operational
- ✂ Random Forest model trained on NSL-KDD dataset for intrusion classification
- ✂ Threats are getting identified correctly.

Expected Outcomes

- ✂ A functional, lightweight IDS deployed on Raspberry Pi for real-time IoT intrusion detection
- ✂ Accurate detection of DoS and Replay attacks using RFID/ESP32-C5-generated traffic

References

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IEEE Format References

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Thank You